

Duration: 2:00 hrs

(Calculator is Allowed)

Max. Marks: 110

Question 1

[Maximum mark: 5]

A teacher records the number of hours, x , that each of eight students spent revising for a test, together with their test score, y , as a percentage. The results are shown in the following table.

x (hours)	2	3	5	6	8	9	11	12
y %	45	52	58	63	71	74	82	88

- (a) Find the value of Pearson's product-moment correlation coefficient, r . [2]
- (b) State the equation of the regression line of y on x . [1]
- (c) (i) Use your regression line to estimate the test score of a student who revised for 10 hours. [2]
- (ii) Comment on the reliability of this estimate.

Answer: (a) Using ti-nspire CX ii

After entering values in table: click → menu → 4 statistics → 1 stat calc → 3 Linear Regression

Using Casio CG-50

After entering values in two lists:
click → F_2 (calc) → F_3 (REG) → $F_1(x)$ → $F_1(ax+b)$

$$r = 0.997$$

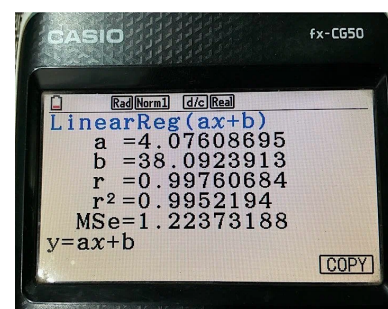
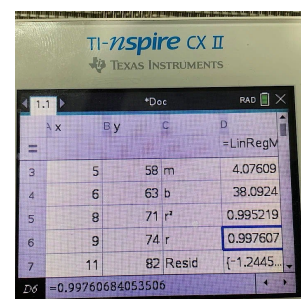
(b) $y = 4.076 \dots x + 38.09 \dots$

(c) (i) $x = 10$

$$y = 4.076 \dots (10) + 38.09 \dots \approx 78.9\%$$

- (ii) The estimate is reliable, because $x = 10$ lies within the range of the data and r is very close to 1 (strong correlation).

interpolation



[Maximum mark: 5]

Find the following

(a) the coefficient of x^4 in $(3 + 2x)^5$; [2](b) hence find the coefficient of x^4 in $(6 - x)(3 + 2x)^5$. [3]**Answer:**

$$(a) (3 + 2x)^5 = {}^5C_0(3)^5 + {}^5C_1(3)^4(2x)^1 + {}^5C_2(3)^3(2x)^2 + {}^5C_3(3)^2(2x)^3 + {}^5C_4(3)^1(2x)^4 + {}^5C_5(2x)^5$$

$$= 243 + 810x + 1080x^2 + 720x^3 + 240x^4 + 243$$

$$\therefore \text{Coefficient of } x^4 = 240$$

$$(b) (6 - x)(243 + 810x + 1080x^2 + 720x^3 + 240x^4 + 243)$$

$$\text{Coeff. of } x^4 = (-1)720 + 6(240) = 720$$

[Maximum mark: 5]

The discrete random variable X has the probability distribution shown in the table, where $a, b \in \mathbb{Q}^+$.

x	-5	1	3	7
$P(X = x)$	a	0.20	b	0.10

Total prob = 1
each $0 \leq P(X=x) \leq 1$

(a) Write down an equation relating a and b .

[1]

The probability distribution shown in the table represents a fair game.

(b) Find the value of a and the value of b .

[4]

Answer:

(a) We know that total probability is 1

$$\begin{aligned}\therefore a + 0.20 + b + 0.10 &= 1 \\ a + b &= 0.70 \quad \text{--- (i)}\end{aligned}$$

(b) Given that the game is fair.

$$\begin{aligned}\therefore E(X) &= 0 \quad \longrightarrow \text{(Mean of distribution is } E(X) \text{ or } \mu) \\ \sum_{i=1}^4 x_i p_i &= 0 \quad \text{(Game is fair)} \\ -5a + 0.20 + 3b + 0.70 &= 0 \\ -5a + 3b &= -0.90 \quad \text{--- (ii)}\end{aligned}$$

From part(a),

$$\begin{aligned}b &= (0.70 - a) \\ \therefore -5a + 3(0.70 - a) &= -0.90 \\ -8a &= -3.0 \\ a &= \frac{3}{8} = 0.375 \\ \therefore b &= 0.70 - \frac{3}{8} = 0.325\end{aligned}$$

[Maximum mark: 8]

During a chemical reaction, the rate at which a compound is produced, in grams per minute, is modelled by

$$r(t) = 12te^{-0.5t}$$

where t is the time in minutes after the start of the reaction.

- (a) Find an expression for $r'(t)$. [3]
- (b) Hence find the time at which the rate of production is decreasing most rapidly. [3]
- (c) Find the total mass of compound produced during the first 5 minutes. [2]

Answer:

(a) $r(t) = 12te^{-0.5t}$

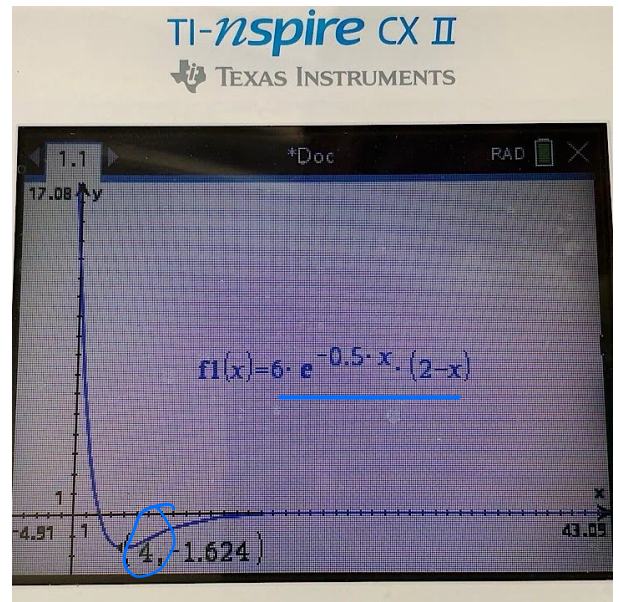
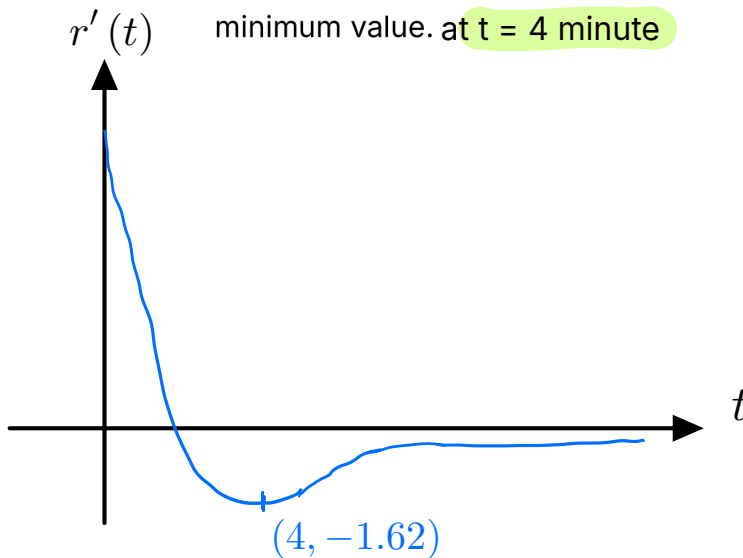
using product rule, we get

$$((uv)') = u'v + uv'$$

$$r'(t) = 12(1)e^{-0.5t} + 12t\left(e^{-0.5t} \times \frac{-1}{2}\right) = 12e^{-0.5t} - 6te^{-0.5t}$$

$$= 6e^{-0.5t}(2 - t)$$

- (b) The rate of production decreases most rapidly when $r'(t)$ takes its minimum value. at $t = 4$ minute



- (c) (menu → Analyze Graph → Minimum)

$$\text{Total mass produced} = \int_0^5 10te^{-2t} dt \approx 2.50 \text{ g}$$

$f(x) = a \sin(bt) + k \rightarrow b = \frac{2\pi}{\text{Period}}$ or $b = \frac{360}{\text{Period}}$.
 amplitude = $|a|$

In a harbour is modelled by

$d(t) = a \sin(bt) + k \quad 0 \leq t \leq 24$



[Maximum mark: 6]

The depth of water, d metres, in a harbour is modelled by

$$d(t) = a \sin(bt) + k, \quad 0 \leq t \leq 24,$$

where t is the time in hours after midnight and a, b, k are positive constants.

On a particular day the first high tide, of depth 8.5 m, occurs at $t = 3$, and the following low tide, of depth 3.5 m, occurs at $t = 9$

- (a) (i) Find the value of a .
(ii) Find the value of k . [2]
- (b) Show that $b = \frac{\pi}{6}$. [1]
- (c) Find the first time after midnight when the depth of water is exactly 7 metres. Give your answer to the nearest minute. [3]

Answer:

$$(a) \quad a = \frac{\text{Maximum} - \text{Minimum}}{2} = \frac{8.5 - 3.5}{2} = 2.5$$

$$(ii) \ k = \frac{Maximum + Minimum}{2} = \frac{8.5 + 3.5}{2} = 6$$

- (b) Maximum at $t = 3$ and Minimum at $t = 9$

$$\therefore \text{Period} = 2 \times (9 - 3) = \underline{\underline{12}}$$

$$\therefore b = \frac{2\pi}{Period} = \frac{2\pi}{12} = \frac{\pi}{6}$$

(c) $d(t) = 2.5 \sin\left(\frac{\pi}{6}t\right) + 6$

From the graph of the sign diagram, we get

$$6 + 2.5 \sin \left(\frac{\pi}{6} t \right) = 7$$

$$\sin\left(\frac{\pi}{6}t\right) = 0.4$$

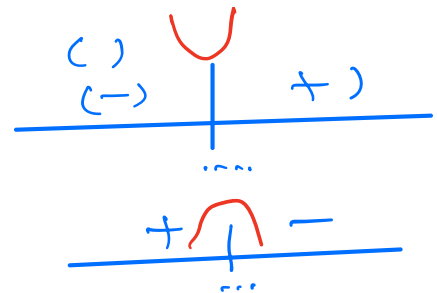
$$t \approx 0.7859 \dots$$

The water is at a depth of 7 metres at 12:47 am

Max to Min



Period = 2 x (half Period)



(converted to minutes)

$$0.7859 \dots \times 60 \approx 47.156$$

Question 6

[Maximum mark: 6]

The 27 staff members at a high school are lining up for annual staff photos.

They will be arranged in 3 rows.

- Front row: 7 staff members
- Middle row: 9 staff members
- Back row: 11 staff members

For the first photo, the staff are arranged so that the 11 tallest staff members are in the back row, and the 7 shortest staff members are in the front row.

(a) Find the number of possible arrangements for the first photo. [2]

For the second photo, the 5 members of senior management will occupy the middle 5 positions in the front row, and there will be no restrictions on the other positions.

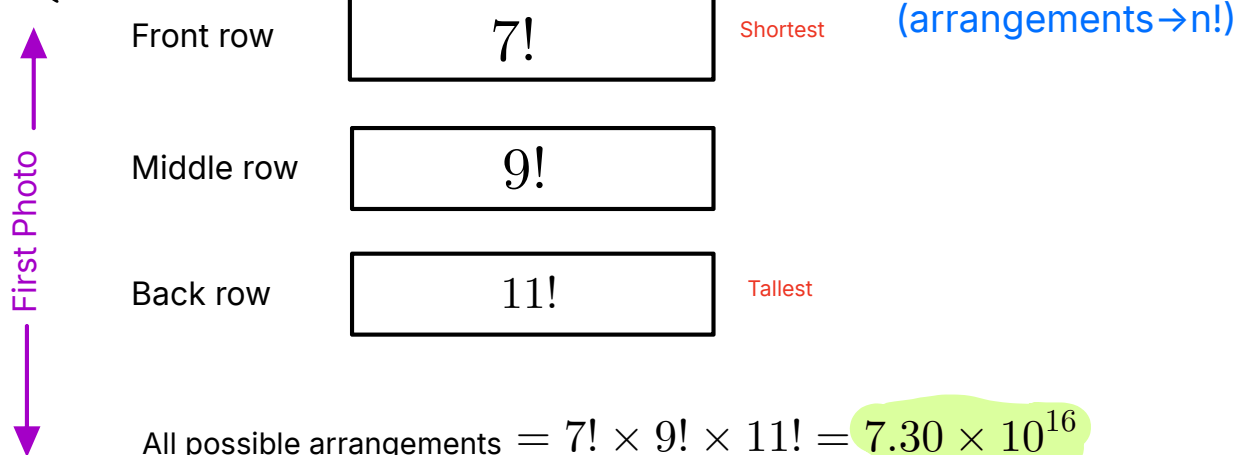
(b) Find the number of possible arrangements for the second photo. [2]

For the third photo, the 5 members of senior management are not involved, and the remaining staff are to be arranged in two rows, 10 in the front row and 12 in the back row.

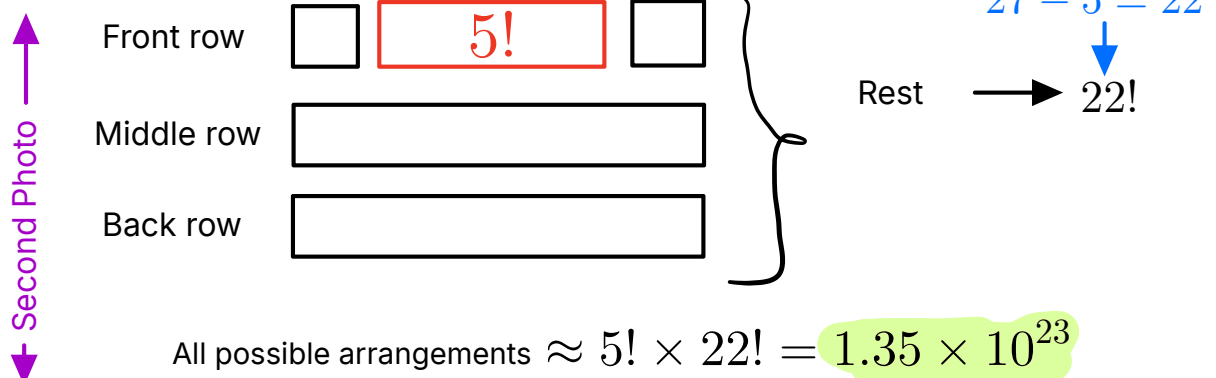
For this photo, the members of each department will stand together, with the 5 mathematics staff and 7 science staff in the back row, and the 6 humanities staff and 4 languages staff in the front row.

(c) Find the number of possible arrangements for the third photo. [2]

Answer: (a)



(b)



(c)

For the third photo, the 5 members of senior management are not involved, and the remaining staff are to be arranged in two rows, 10 in the front row and 12 in the back row.

For this photo, the members of each department will stand together, with the 5 mathematics staff and 7 science staff in the back row, and the 6 humanities staff and 4 languages staff in the front row.



Third Photo

Front row(10)

$$6! \times 4! \times 2$$

Human(6)+Lang(4)

Back row(12)

$$5! \times 7! \times 2$$

Math(5)+Sci(7)

$$27 - 5 = 22$$

↓
22

All possible arrangements are:

$$(6! \times 4! \times 2) \times (5! \times 7! \times 2) \approx 4.18 \times 10^{10}$$

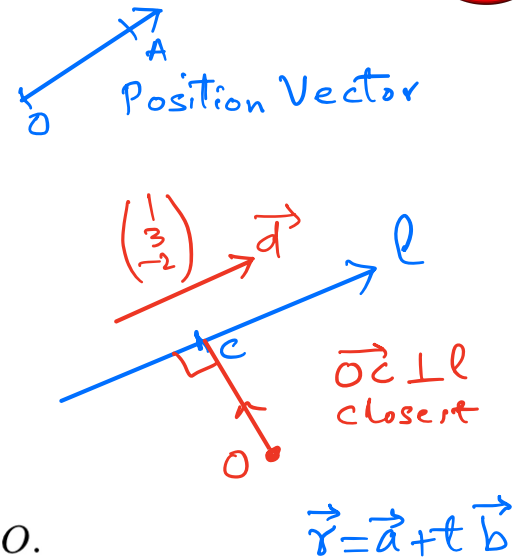
[Maximum mark: 8]

Points A and B have position vectors

$$\vec{OA} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \quad \vec{OB} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}.$$

The point C lies on the line l with equation

$$\mathbf{r} = \begin{pmatrix} -2 \\ -4 \\ 7 \end{pmatrix} + t \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}, \quad t \in \mathbb{R},$$

and C is the point on l that is closest to the origin O .(a) Find the position vector of C .

[3]

(b) Hence find the area of triangle ABC .

[5]

Answer:(a) C lies on line l , so direction vector & position vector of C is

$$\vec{d} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} \quad \text{and} \quad \vec{OC} = \begin{pmatrix} -2+t \\ -4+3t \\ 7-2t \end{pmatrix}$$

Now point C is closest to origin, so vector $\vec{OC}$ is perpendicular to line l and so is perpendicular to direction vector $\vec{d}$.

$$\therefore \vec{d} \cdot \vec{OC} = 0$$

$$\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2+t \\ -4+3t \\ 7-2t \end{pmatrix} = 0$$

$$(1)(-2+t) + (3)(-4+3t) + (-2)(7-2t) = 0$$

$$-28 + 14t = 0$$

$$\therefore t = 2$$

$$\left(\begin{array}{l} \text{dot-product formula 3.13 is} \\ \mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3 \end{array} \right)$$

Hence position vector



$$\vec{OC} = \begin{pmatrix} -2+2 \\ -4+3(2) \\ 7-2(2) \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$$

(b)

$$\vec{OA} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad \vec{OB} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \quad \vec{OC} = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$$

$$\vec{AB} = \begin{pmatrix} 3 & -1 \\ 1 & -0 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \quad \vec{AC} = \begin{pmatrix} 0 & -1 \\ 2 & -0 \\ 3 & -2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 2 & 1 & -1 \\ -1 & 2 & 1 \end{vmatrix} = 3i - j + 5k \quad \left(\mathbf{v} \times \mathbf{w} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix} \right) \quad \text{3.16}$$

$$\begin{aligned} \therefore \text{area of } \triangle ABC &= \frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} \sqrt{3^2 + (-1)^2 + (5)^2} \\ &= \frac{1}{2} \sqrt{35} \end{aligned}$$

$$\therefore \text{area of } \triangle ABC \approx 2.96 \text{ units}^2$$

[Maximum mark: 6]

Air is pumped into a spherical balloon at a constant rate of $40\text{cm}^3\text{s}^{-1}$.

(a) Find the rate of increase of the radius when the surface area is 800cm^2 . [3]

(b) Hence find the rate of increase of the surface area at the same moment. [3]

Answer:

(a) According to formula 3.1 in formula booklet, volume of sphere is

$$V = \frac{4}{3}\pi r^3$$

$$\therefore \frac{dV}{dr} = 4\pi r^2$$

Given that

$$\frac{dV}{dt} = 40\text{cm}^3\text{s}^{-1}$$

Using chain rule

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$\therefore 40 = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$40 = 4\pi r^2 \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = \frac{40}{4\pi r^2}$$

We have surface area $4\pi r^2 = 800$

$$\therefore \frac{dr}{dt} = \frac{40}{800} = \frac{1}{20} \approx 0.05\text{cm s}^{-1}$$

(b) $s = 4\pi r^2 \Rightarrow 800 = 4\pi r^2 \Rightarrow r = 7.9788\dots$

$$\frac{ds}{dr} = 8\pi r \quad \frac{ds}{dt} = \frac{ds}{dr} \times \frac{dr}{dt}$$

$$\therefore \frac{ds}{dt} = 8\pi r \times \frac{1}{20} = 8\pi (7.9788\dots) \times \frac{1}{20} \approx 10.0\text{cm}^2\text{s}^{-1}$$

Therefore, surface area is increasing at the rate of $10.0\text{cm}^2\text{s}^{-1}$

[Maximum mark: 8]

(a) Show that the equation

$$\sec^2 \theta - \operatorname{cosec}^2 \theta = \tan 2\theta$$

Can be written as

$$\tan^6 \theta - \tan^4 \theta + 2 \tan^3 \theta - \tan^2 \theta + 1 = 0$$

[3]

(b) Hence, solve the equation $\sec^2 \theta - \operatorname{cosec}^2 \theta = \tan 2\theta$ for $\theta \in [0, 2\pi]$.

[3]

The equation $\sec^2 \theta - \operatorname{cosec}^2 \theta = \tan 2\theta$ has one solution for $\theta \in (\pi, a)$.(c) Find the maximum value of a .

[2]

Answer:

$$(a) \quad \sec^2 \theta - \operatorname{cosec}^2 \theta = \tan 2\theta$$

$$(1 + \tan^2 \theta) - (1 + \cot^2 \theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

(formula: 3.9, 3.10)

$$\cancel{1} + \tan^2 \theta - \cancel{1} - \left(\frac{1}{\tan^2 \theta} \right) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$(1 - \tan^2 \theta) (\tan^4 \theta - 1) = 2 \tan \theta (\tan^2 \theta)$$

$$\tan^4 \theta - 1 - \tan^6 \theta + \tan^2 \theta = 2 \tan^3 \theta$$

$$-\tan^6 \theta + \tan^4 \theta - 2 \tan^3 \theta + \tan^2 \theta - 1 = 0$$

$$\tan^6 \theta - \tan^4 \theta + 2 \tan^3 \theta - \tan^2 \theta + 1 = 0$$

(b)

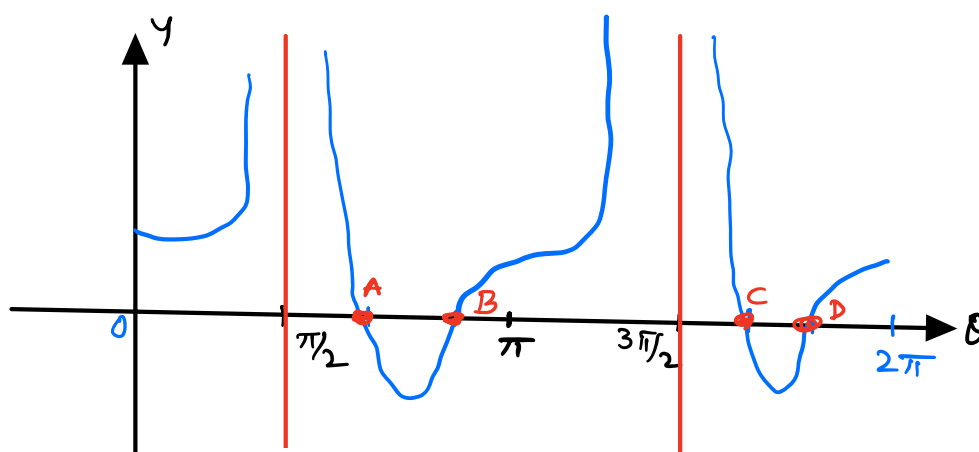
(Using G- D. C.)

$$A (2.134, 0)$$

$$B (2.578, 0)$$

$$C (5.276, 0)$$

$$D (5.720, 0)$$



$$\therefore \theta = 2.13, 2.58, 5.28, 5.72$$

(C.) From part(a), The two solutions greater than π are

$$\theta = 5.276 \dots, 5.720 \dots$$

Therefore, the maximum value of a is

$$a = 5.72$$

SECTION B



Question 10

[Maximum mark: 14]

A piston moves back and forth along a straight line inside an engine. Its velocity, $v \text{ m.s}^{-1}$, at time t seconds is modelled by

$$v_1(t) = 5 \cos(t) - 2, \quad 0 \leq t \leq 8,$$

where t is measured in radians.

(a) Determine the velocity of the piston when $t = 0$. [2]

(b) Find the values of t in the interval $0 \leq t \leq 8$ for which the piston is momentarily at rest. [3]

(c) Find the acceleration of the piston when $t = 3$. [2]

The piston passes through the point O when $t = 0$.

(d) Find the total distance travelled by the piston during the first 4 seconds. [3]

A second piston moves with a similar motion to the first piston, with velocity, $v \text{ m.s}^{-1}$, at time t seconds modelled by

$$v_2(t) = 5 \cos(t - 1) - 2, \quad 0 \leq t \leq 8,$$

where t is measured in radians.

(e) Determine how much further the first piston has travelled than the second piston, at the first time the second piston is momentarily at rest. [4]

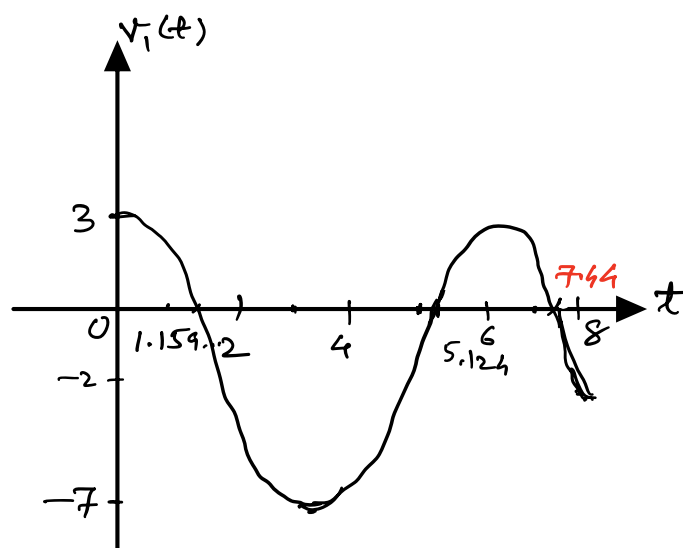
Answer:

(a) $V_1(0) = 5 \cos(0) - 2 = 5 - 2 = 3 \text{ m.s}^{-1}$

(b) Using GDC, over $0 \leq t \leq 8$

Piston is momentarily at rest for

$t = 1.165, t = 5.125, t = 7.445$



(c) $a(t) = v_1'(t) = -5 \sin t$

$\therefore a(3) = -5 \sin(3) \approx -0.706 \text{ m.s}^{-2}$

$(v_1(t) = 5 \cos(t) - 2)$

(d) $d = \int_0^4 |(5 \cos t - 2)| dt \approx 16.3m$

(e) A second piston moves with a similar motion to the first piston, with velocity, v m.s⁻¹, at time t seconds modelled by

$$v_2(t) = 5 \cos(t - 1) - 2, \quad 0 \leq t \leq 8,$$

where t is measured in radians.

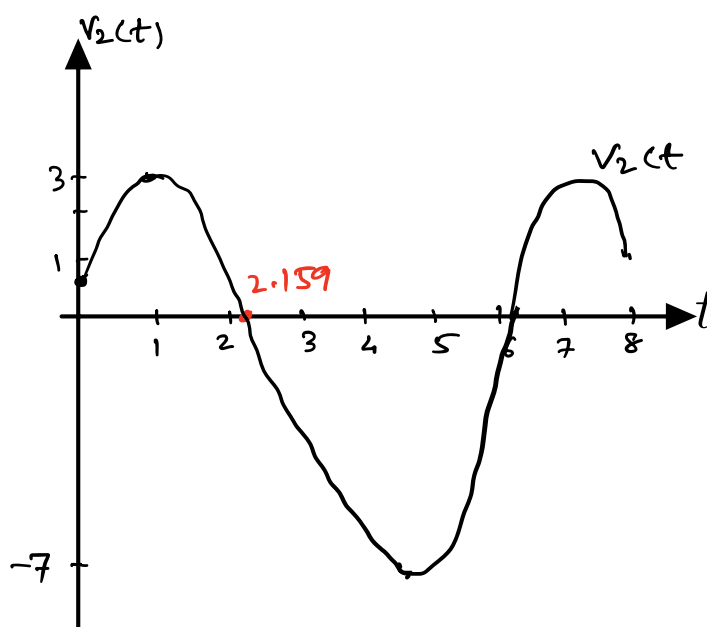
(e) Determine how much further the first piston has travelled than the second piston, at the first time the second piston is momentarily at rest.

[4]

Using GDC, over

$$t = 2.159 \dots s$$

When $v_2(t)$ is first at rest



$$\text{difference} = \int_0^{2.159\dots} |5 \cos(t) - 2| dt - \int_0^{2.159\dots} |5 \cos(t - 1) - 2| dt$$

$$= 0.2163 \dots$$

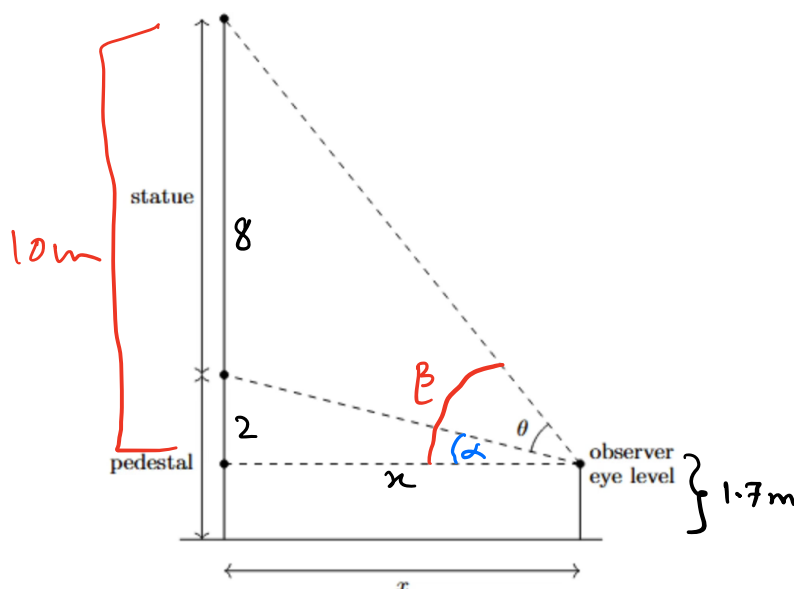
$$\therefore \text{difference} = 0.216m$$

Question 11

[Maximum mark: 19]

A statue of height 8 m stands on a pedestal of height 3.7 m, on horizontal ground. An observer, whose eye-level is at 1.7 m, stands at a horizontal distance x metres from the base of the pedestal, where $x > 0$.

Let θ be the angle subtended at the observer's eye by the statue, as shown in the diagram.



- Show that $\theta = \arctan\left(\frac{8x}{x^2 + 20}\right)$ [4]
- Hence show that $\frac{d\theta}{dx} = \frac{8(20 - x^2)}{(x^2 + 4)(x^2 + 100)}$ [5]
- Find x such that $\frac{d\theta}{dx} = 0$. [2]
- State whether this value of x gives a maximum or minimum value of θ . Justify your answer. [2]
- Find the maximum value of θ , in degrees. [2]

A second observer is standing directly between the statue and the first observer. The angle being subtended by the statue to the eye of both observers is 0.5 radians.

It can be assumed that the eye-level of both observers is at 1.7 m, and that the two observers are not standing in the same spot.

- Determine the distance between the two observers. [4]

Answer:

$$(a) \quad \theta = \beta - \alpha$$

$$\tan(\beta) = \frac{10}{x}$$

$$\therefore \tan \theta = \tan (\beta - \alpha)$$

(3.10)

$$\left(\tan (A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} \right)$$

$$\tan \theta = \frac{\tan \beta - \tan \alpha}{1 + \tan \beta \cdot \tan \alpha}$$

$$= \frac{\left(\frac{10}{x} \right) - \left(\frac{2}{x} \right)}{1 + \left(\frac{10}{x} \right) \cdot \left(\frac{2}{x} \right)}$$

$$\tan \theta = \frac{\left(\frac{8}{x} \right)}{\left(\frac{x^2 + 20}{x^2} \right)} = \frac{8x}{x^2 + 20}$$

$$\therefore \theta = \tan^{-1} \left(\frac{8x}{x^2 + 20} \right)$$

(b) Using $\frac{d}{dx} (\tan^{-1} x)$ (5.15) and chain rule (5.6)

$$\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{8x}{x^2 + 20} \right)^2} \times \frac{d}{dx} \left(\frac{8x}{x^2 + 20} \right)$$

$$\left(\begin{array}{l} \frac{d}{dx} (\tan^{-1} y) = \frac{1}{1 + y^2} \\ y = \frac{8x}{x^2 + 20} \end{array} \right)$$

$$= \frac{(x^2 + 20)^2}{(x^2 + 20)^2 + 64x^2} \times \frac{8(x^2 + 20) - 8x(2x)}{(x^2 + 20)^2}$$

$$\frac{d\theta}{dx} = \frac{160 - 8x^2}{x^4 + 104x^2 + 400}$$

$$\frac{d\theta}{dx} = \frac{8(20 - x^2)}{(x^2 + 4)(x^2 + 100)}$$

(As required)

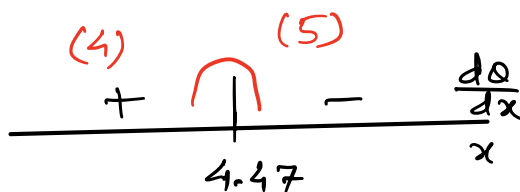
$$\left(\begin{array}{l} x^4 + 4x^2 + 100x^2 + 400 \\ = x^2(x^2 + 4) + 100(x^2 + 4) \\ = (x^2 + 4)(x^2 + 100) \end{array} \right)$$

(c) $\frac{d\theta}{dx} = 0 \Rightarrow 8(20 - x^2) = 0$

$$\therefore x^2 = 20 \Rightarrow x = 4.47$$

(d) $\frac{d\theta}{dx} \Big|_{x=4} > 0$

and $\frac{d\theta}{dx} \Big|_{x=5} < 0$



As the gradient function changes from positive to negative

$\therefore x = 4.47$ gives Maximum value of θ

The angle subtended by statute is maximum when observer stands at 4.47 metres away

(e) (e) Find the maximum value of θ , in degrees.

From part (a)

$$\begin{aligned}\theta &= \tan^{-1} \left(\frac{8x}{x^2 + 20} \right) \\ &= \tan^{-1} \left(\frac{8(4.47)}{(4.47)^2 + 20} \right)\end{aligned}$$

$$\theta = 41.8^\circ$$

(f)

A second observer is standing directly between the statue and the first observer. The angle being subtended by the statue to the eye of both observers is 0.5 radians.

It can be assumed that the eye-level of both observers is at 1.7 m, and that the two observers are not standing in the same spot.

(f) Determine the distance between the two observers.

[4]

Let first observe is at C and
second observe is at B

From part (a)

$$\begin{aligned}\theta &= \tan^{-1} \left(\frac{8x}{x^2 + 20} \right) \\ 0.5 &= \tan^{-1} \left(\frac{8x}{x^2 + 20} \right) \\ \tan(0.5) &= \frac{8x}{x^2 + 20}\end{aligned}$$

$$x^2 (\tan 0.5) - 8x + 20 \tan 0.5 = 0$$

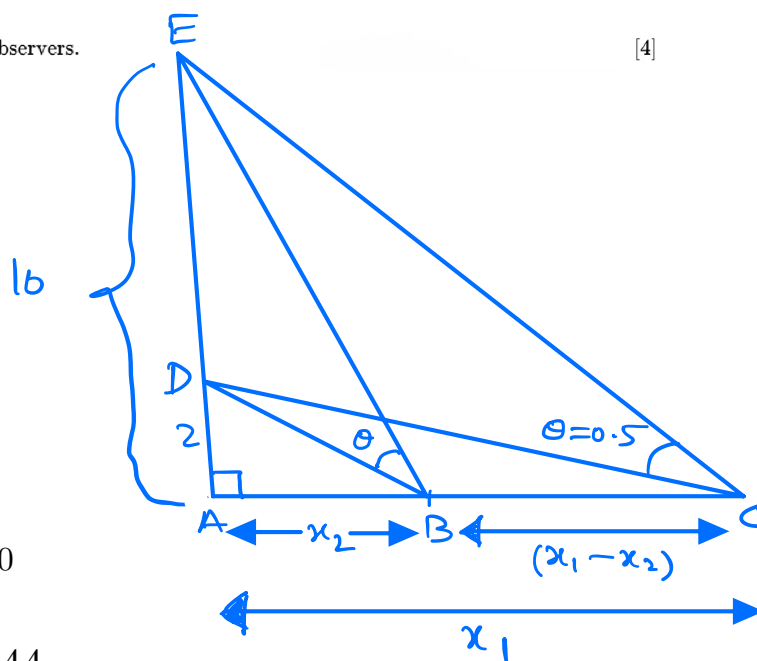
Using GDC

$$x_1 = 13.119, x_2 = 1.5244$$

$$x_1 - x_2 = 11.59$$

Hence distance between observers is

11.6m



[Maximum mark: 20]

A workshop employs three mechanics: Anna, Brian and Christine.

The time taken, X minutes, for Anna to complete a repair is normally distributed with mean μ and standard deviation σ . It is known that

$$P(X < 20) = 0.1 \quad \text{and} \quad P(X > 35) = 0.2.$$

- (a) Find the value of μ and the value of σ . [4]

The workshop charges \$120 per hour for Anna's work, plus a flat fee of \$100. The amount charged for a repair carried out by Anna is \$Y.

- (b) Find the distribution of Y , stating its parameters. [3]

A repair is classified as **expensive** if the amount charged is more than \$180.

- (c) Hence find the probability that a repair carried out by Anna is classified as expensive. [2]

Anna carries out 12 repairs in a week, which may be assumed to be independent.

- (d) Given that at least 3 of these repairs are classified as expensive, find the probability that exactly 3 are classified as expensive. [3]

The time taken, T minutes, for Brian to complete a repair is normally distributed with mean 25 and standard deviation 5. The workshop charges \$r per hour for Brian's work, plus the same flat fee of \$100.

A repair carried out by Brian is twice as likely to be classified as expensive as a repair carried out by Anna.

- (e) Find the value of r . [4]

In a typical week, Anna carries out 40% of all repairs, Brian carries out 35% and Christine carries out 25%. A repair carried out by Christine is classified as expensive with probability 0.2.

- (f) Given that a repair was classified as expensive, find the probability that it was carried out by Brian. [4]

Answer:

$$(a) \quad X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{x - \mu}{\sigma} \quad \text{Is standardisation equation (4.12)}$$

$$\therefore Z \sim N(0, 1^2)$$

$$P(X < 20) = 0.1 \Rightarrow P\left(z < \frac{20 - \mu}{\sigma}\right) = 0.1$$

Using inverse normal on GDC

$$\frac{20 - \mu}{\sigma} = \text{InvNorm}(0.1) \Rightarrow \frac{20 - \mu}{\sigma} = -1.28155 \dots$$

$$\therefore 20 - \mu = -1.28155 \dots \sigma \quad \text{--- (i)}$$

and

$$P(X > 35) = 0.2 \Rightarrow P\left(Z > \frac{35 - \mu}{\sigma}\right) = 0.2$$

Using inverse normal on GDC

$$\frac{35 - \mu}{\sigma} = \text{InvNorm}(0.2) \Rightarrow \frac{35 - \mu}{\sigma} = 0.84162 \dots$$

$$\therefore 20 - \mu = -1.28155 \dots \sigma \quad \text{--- (i)}$$

$$\therefore 35 - \mu = 0.84162 \dots \sigma \quad \text{--- (ii)}$$

Solving eq(i) and (ii), Using

$$\sigma = 7.06, \quad \mu = 29.1$$

$$\rightarrow E(x) = \mu$$

$$Var(x) = \sigma^2$$

(b) Anna's rate, as X is measured in minutes,

$$\$120 \text{ per hr} \longrightarrow \frac{120}{60} = \$2 \text{ per minute}$$

$$\therefore Y = 2X + 100$$

Using formula (4.14) for expectation & variance, we have

$$E(Y) = E(aX + b) = aE(X) + b$$

$$E(2X + 100) = 2E(X) + 100$$

$$E(Y) = 2 \times 29.05 \dots + 100 \quad (\mu = 29.05)$$

$$\therefore E(Y) = \underline{158.1 \dots}$$

and

$$Var(Y) = Var(2X + 100) = 2^2 Var(X)$$

$$= 4 \times 7.06^2 \quad (\sigma = 7.06)$$

$$\therefore Var(Y) = 199.6 \dots \approx \underline{200}$$

Since linear transformation of a normally distributed random variable is also normally distributed. Therefore

$$Y \sim N(\underline{158}, \underline{200}^2)$$

(c) we have

$$Y \sim N(158.1 \dots, 199.6 \dots)$$

$$\therefore P(Y > 180) \approx \underline{0.0606}$$

(d)

Anna carries out 12 repairs in a week, which may be assumed to be independent.

(d) Given that at least 3 of these repairs are classified as expensive, find the probability that exactly 3 are classified as expensive.

This is Binomial distribution, with W be the repair work

- $n = 12$ (Fixed trial)
- each trial is independent
- Only two outcomes, **expensive, not expensive**
- Probability of success is constant. $p=0.06064$

$$\therefore W \sim B(12, 0.06064)$$

$$\left(P(A|B) = \frac{P(A \cap B)}{P(B)} \right)$$

$$\begin{aligned} P(W = 3 | W \geq 3) &= \frac{P(W = 3 \cap W \geq 3)}{P(W \geq 3)} = \frac{P(w = 3)}{P(w \geq 3)} \\ &= \frac{0.02793758}{0.03244796} \end{aligned}$$

$$P(W = 3 | W \geq 3) \approx 0.861$$

(e)

The time taken, T minutes, for Brian to complete a repair is normally distributed with mean 25 and standard deviation 5. The workshop charges \$ r per hour for Brian's work, plus the same flat fee of \$100.

A repair carried out by Brian is twice as likely to be classified as expensive as a repair carried out by Anna.

(e) Find the value of r .

$$\begin{aligned} P(\text{expensive} | \text{Brian}) &= 2 \times 0.06064 \dots \\ &= 0.1212 \end{aligned}$$

As hourly rate is per minute, rate of Brian will be $\frac{r}{60}$ per minute

Therefore, amount charged by Brian is $\frac{r}{60}T + 100$

$$\begin{aligned} \frac{r}{60}T + 100 &> 180 \\ \frac{r}{60}T &> 80 \\ T &> \frac{4800}{r} \end{aligned}$$

As $T > \frac{4800}{r}$ Brian's repair is expensive with probability 0.1212

$$\therefore P\left(T > \frac{4800}{r}\right) = 0.1212$$

Using inverse normal on GDC $T \sim N(25, 5^2)$

$$\frac{4800}{r} = 30.845 \dots$$

$$\therefore r = \$156$$

(f)

In a typical week, Anna carries out 40% of all repairs, Brian carries out 35% and Christine carries out 25%. A repair carried out by Christine is classified as expensive with probability 0.2.

(f) Given that a repair was classified as expensive, find the probability that it was carried out by Brian.

[4]

Let $A \rightarrow$ Repair by Anna
 $B \rightarrow$ Repair by Brian
 $C \rightarrow$ Repair by Christine

$$\therefore P(A) = 0.40, P(B) = 0.35, P(C) = 0.25$$

Let $M \rightarrow$ Repair is expensive

$$\text{From part (c)} \rightarrow P(M|A) = 0.06064$$

$$\text{From part (e)} \rightarrow P(M|B) = 0.1212$$

And from information given about Christine

$$P(M|C) = 0.2$$

Using Bayes' theorem formula (4.13)

$$\begin{aligned} P(B|M) &= \frac{P(B)P(M|B)}{P(A)P(M|A) + P(B)P(M|B) + P(C)P(M|C)} \\ &= \frac{0.35 \times 0.1212}{(0.40 \times 0.06064) + (0.35 \times 0.1212) + (0.25 \times 0.2)} \end{aligned}$$

$$\therefore P(B|M) = 0.364$$



Request from teacher:

If you really like my method of teaching, then please like, share and subscribe to my YouTube channel.

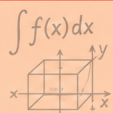
*Also, click bell icon to get my new **LIVE** videos.*

End of Lesson

Contact me on:  **9326363793**

You can download the PDF of this lesson from the link, given in the description of this video. My channel link is clickable at the bottom of the all pages.



$\int f(x)dx$

☒ Past Paper Trends
☒ Important Topics
☒ Exam Style Questions

IB Mathematics

AAHL PAPER-3

PREDICTION EXAM

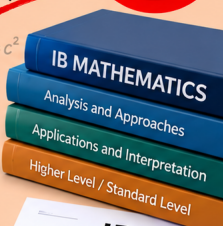
for November 2026

with SOLUTION

$f_i(x)$
 Prepare Practice Perform!
 $a^2 + b^2 = c^2$



 $\sin x$



 IB Mathematics Examination
 Higher Level / Standard Level
 Your Success Our Goal!



 [youtube.com/@IBMATH-in](https://www.youtube.com/@IBMATH-in)

 **9326363793**